

TESSA Boat – Tennessee Environmental Surface Skimming Autonomous Boat – Solar-Powered Autonomous Waterway Cleanup Vessel

Sponsor: Self-directed, no sponsor

Background and Objective

Floating trash and debris are common problems in streams, rivers, ponds, and other waterways. Traditional cleanup methods often require significant amounts of manual labor and may be difficult or expensive to perform in areas that are challenging to access. Around Tennessee all the watershed runoff floats to the Gulf of Mexico/America, contributing to the 8-14 million metric tons of plastic waste that enters the world's oceans every year. The objective of this project is to develop a small, solar-powered boat capable of collecting floating debris from waterways.

Background and Objective

The boat will use a microcontroller as its primary control system and will integrate electric propulsion, sensors, battery storage, and a solar-power system. The autonomous system will be designed to allow the robot to navigate a designated area and systematically collect floating debris. There will also be an option for manual control if necessary.

Project Scope

Research and compare different hull, propulsion, and debris-collection designs. Evaluate different motor and underwater-thruster systems based on thrust, voltage, efficiency, and controllability. Design the electrical power system, including solar panels, battery storage, voltage regulation, and integrate a microcontroller as the primary control system.

Develop a motor/thruster control system capable of movement in all direction including forward, reverse, turning, and stopping. Develop both manual remote-control, potential App interface and autonomous operating modes. Develop a waterproof enclosure for the electronics to be safely integrated into the hull design.

Deliverables

Fully functional prototype of the solar-powered waterway-cleanup boat. Demonstration of manual and autonomous operation for floating-debris collection. Demonstration of solar-power and battery-storage system, microcontroller-based control system, autonomous navigation system, mechanical CAD design for hull and component storage and final Senior Design report and presentation.

Expected Project Outcome

The final product will be a small-scale autonomous boat capable of collecting floating debris while using solar energy to supplement its onboard battery system. Prototype will demonstrate the feasibility of using embedded electronics, renewable energy, possible autonomous and app control, and robotic propulsion for environmental cleanup applications.